

Result Tables

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Main results

Each column is a held-out target. each row is a different method (Source-only = no adaptation at all. BN-only = no-op control since our encoder has no BatchNorm stats. Tent = a known TTA baseline that collapses on 2 targets. Self-training only = ours without consistency. Ours = full method. Oracle = supervised upper bound). Ours wins on 3 targets out of 4; each cell is EER% / AUC, lower/higher is better.

Method	ASVspoof2019	LibriSpeech-TTS	In-the-Wild	Arabic
Source-only	4.76 / .990	34.36 / .720	12.57 / .944	22.59 / .854
BN-only (no-op)	4.76 / .990	34.36 / .720	12.57 / .944	22.59 / .854
Tent	3.70 / .987	34.74 / .690	26.53 / .743	47.66 / .524
Self-training only	4.07 / .987	33.79 / .708	11.63 / .947	22.05 / .855
Ours (full TTA)	3.69 / .994	34.38 / .722	10.83 / .960	20.74 / .873
Oracle (supervised upper bound)	0.70 / 1.000	12.54 / .950	3.65 / .994	9.14 / .970

Gain decomposition

Splits the total EER change into two steps: Δ_1 = effect of adding self-training alone, Δ_2 = effect of then adding consistency on top. Δ_2 is negative (helps) on 3 targets but +0.59 on LibriSpeech-TTS

Target	Source EER	Δ_1 (self-train)	Δ_2 (consistency)
ASVspoof2019	4.76	-0.69	-0.39
LibriSpeech-TTS	34.36	-0.57	+0.59
In-the-Wild	12.57	-0.94	-0.79
Arabic	22.59	-0.54	-1.31

Backbone comparison

XLS-R = our SSL-pretrained encoder. RawNet2Lite = same epochs, same data, but trained from random weights (no SSL pretrain). RawNet2Lite collapses everywhere

Target	XLS-R (ours)	RawNet2Lite (no SSL)
ASVspoof2019	4.76 / .990	47.47 / .539
LibriSpeech-TTS	34.36 / .720	51.14 / .483
In-the-Wild	12.57 / .944	39.35 / .649
Arabic	22.59 / .854	57.01 / .411

Comparison to train-time domain-adversarial training (DANN)

DANN sees the target’s unlabeled audio during training (more access than we get) yet is worse than Source-only on 2 targets and barely better on the others. Ours beats DANN on every target at $\sim 1/3$ the compute (9 vs. 30 min per target-seed).

Target	Source	DANN (30 min)	Ours (9 min)
ASVspoof2019	4.76 / .990	6.61 / .980	3.69 / .994
LibriSpeech-TTS	34.36 / .720	34.81 / .724	34.38 / .722
In-the-Wild	12.57 / .944	11.95 / .931	10.83 / .960
Arabic	22.59 / .854	29.51 / .724	20.74 / .873

Inductive check

Rules out “the model just memorised the scored clips”: adapt on one half of the target, score on the disjoint other half. The main table’s pattern replicates almost exactly (wins on 3 out of 4), so the gain is real, not memorisation.

Target	Source (half)	Ours (disjoint half)
ASVspoof2019	4.93	4.16
LibriSpeech-TTS	34.49	34.70
In-the-Wild	12.66	11.13
Arabic	22.29	21.14

Domain divergence before/after adaptation

$\hat{d}_{\mathcal{A}}$ and domain-probe accuracy both measure how easily a linear classifier can tell source from target embeddings apart; both go *up* slightly after adaptation on every target, showing adaptation works by re-shaping the decision boundary, not by erasing domain identity (which is what DANN tries to do, and underperforms at) and i tries DANN but failed.

Target	$\hat{d}_{\mathcal{A}}$		Domain probe acc. (%)	
	source	adapted	source	adapted
ASVspoof2019	1.572	1.609	86.6	89.9
LibriSpeech-TTS	1.460	1.504	86.4	86.6
In-the-Wild	1.335	1.357	80.4	81.6
Arabic	1.545	1.606	87.9	89.1

Per-language MLAAD fake-recall

Fake-clip recall averaged over 32 languages seen in training vs. 6 languages held out of every source pool entirely. The 0.6-point gap is negligible – unseen languages are detected almost as well as seen ones.

Language group	Mean fake-clip recall
Seen in source training (32 langs)	93.9%
Held out entirely (6 langs)	93.3%